4.1 Sensing and Estimation

板载状态估计问题 ability to estimate their state, the position, the orientation and the velocity, and they must be able to do on board

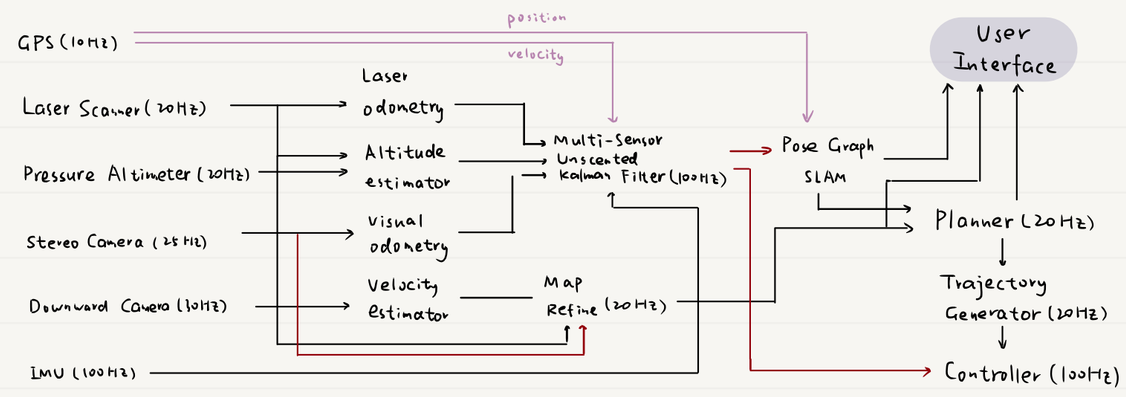
IMU提供的信息: Information about the absolute attitude (orientation) of the robot

Information (via integration) about the relative motion from x_{i-1} to x_i

Laser 提供的信息. Information about the bearing of features relative to the current robot position

Information about the depth of features relative to the current robot position

Estimation and Control Architecture



4.2 Nonlinear Control

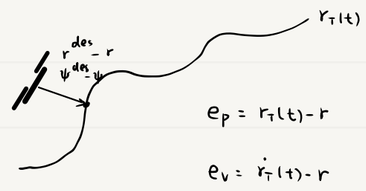
能够在远离平衡点的位置控制 robot.

Trajectory Tracking

Given $r_T(t)$ $\dot{r}_T(t)$ $\ddot{r}_T(t)$

$r^{des}(t)$ $\dot{r}^{des}(t)$, $\ddot{r}^{des}(t)$

$\psi^{des}(t)$, $\dot{\psi}^{des}(t)$, $\ddot{\psi}^{des}(t)$



$e_p = r_T(t) - r$

$e_v = \dot{r}_T(t) - r$

want $(\ddot{r}_T(t) - \ddot{r}_c) + k_d e_v + k_p e_p = 0$

command acceleration calculated by the controller

$$r_T(t) = \begin{bmatrix} x^{des}(t) \\ y^{des}(t) \\ z^{des}(t) \\ \psi^{des}(t) \end{bmatrix}$$

$$u_1 = (\ddot{r}^{des} + k_v \dot{e}_r + k_p e_r + mg a_3) \cdot R b_3$$

$$R^{des} b_3 = \frac{t}{\|t\|} \quad \psi = \psi^{des} \quad e_R(R^{des}, R)$$

$$u_2 = \omega \times I\omega + I(-k_R e_R - k_\omega e_\omega)$$

How to determine R^{des}?

$R^{des} b_3 = \frac{t}{\|t\|}$ $\vec{b}_3$ 与 t 对齐

$\psi = \psi^{des}$

How to calculate the error $e_R(R^{des}, R)$

$R \to R^{des}$

$\Delta R = R^T R^{des}$

Stability

large basin of attraction

从哪些旋转角度开始不可能收敛到所需角度

$tr[I - (R^{des})^T R] < 2$

$$\|e_\omega(0)\|^2 \leq \frac{2}{\lambda_{min}(I)} k_R \left(1 - \frac{1}{2} tr[I - (R^{des})^T R]\right)$$

more stable

Sequential Composition

线性化悬停控制器、旋转控制器、姿态控制器

Trajectory Planning

Inputs u_1, u_2 $\quad u_1 = \sum_{i=1}^{4} F_i \quad u_2 = L \begin{bmatrix} 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ \mu & -\mu & \mu & -\mu \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix}$

所有状态变量和输入都可以写成平面输出及其导数的平滑函数.

State $(q, \dot{q})$ $\begin{bmatrix} \vec{r} \\ \vec{v} \\ \vec{a} \\ \vec{j} \\ \vec{s} \\ \psi \\ \dot{\psi} \\ \ddot{\psi} \end{bmatrix} \longleftrightarrow \begin{bmatrix} \vec{q} \\ \dot{\vec{q}} \\ u_1 \\ \dot{u}_1 \\ \ddot{u}_1 \\ \ddot{\vec{q}} \end{bmatrix}$

$u_1 = m(a_3 - g \cdot \vec{b}_3)$

$\dot{u}_1 = m j_3$

$\ddot{u}_1 = m s_3 + u_1(q^2 + rp)$

$p = \frac{-m j_2}{u_1} \quad q = \frac{m j_1}{u_1}$

↳ [position, velocity, acceleration, jerk, snap, yaw angle]

Planner Quadrotor

Inputs u_1, u_2 $\quad u_1 = F_2 + F_4 \quad \vec{q} = \begin{bmatrix} y \\ z \\ \phi \end{bmatrix}$

$u_2 = (F_2 - F_4) L$

State $(q, \dot{q})$ Equations of motion

$$\begin{bmatrix} \ddot{y} \\ \ddot{z} \\ \ddot{\phi} \end{bmatrix} = \begin{bmatrix} 0 \\ -g \\ 0 \end{bmatrix} + \begin{bmatrix} -\frac{1}{m}\sin\phi & 0 \\ \frac{1}{m}\cos\phi & 0 \\ 0 & \frac{1}{I_{xx}} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

Differential Flatness

两个向量间的平滑映射 All state variables and the inputs can be written as smooth fns of flat outputs and their derivatives

$$\begin{bmatrix} y \\ z \\ \phi \\ \dot{y} \\ \dot{z} \\ \dot{\phi} \\ \ddot{y} \\ \ddot{z} \\ \ddot{\phi} \\ \dots \end{bmatrix} \rightleftarrows \begin{bmatrix} y \\ z \\ \phi \\ \dot{y} \\ \dot{z} \\ \dot{\phi} \\ u_1 \\ u_2 \\ \dot{u}_1 \\ \dot{u}_2 \end{bmatrix}$$

Diffeomorphism 差异同态

4.3 Control of Multiple Robots

- Each individual acts independently
- Actions are based on local information
- Anonymity in coordination

- Assignments of Goals and Collision Free Trajectories
- Concurrent Assignment and Planning of Trajectories: CAPT
  - Compute optimal trajectories for each assignment (n robots and n goals → $O(n^2)$)
  - Find the assignment with the lowest cost ($O(n^3)$)
- Leader-Follower Network
- Anonymity 匿名编队